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Ergosterol peroxide from Chaga mushroom *(Inonotus Obliquus)* exhibits anti-cancer activity by down-regulation of the *β-catenin* pathway in colorectal cancer

Ju-Hee Kang[1,2,6], Jeong-Eun Jang[1], Siddhartha Kumar Mishra[3], Hee-Ju Lee[4], Chu Won Nho[4], Dongyun Shin[1], Mirim Jin[5], Mi Kyung Kim[2], Changsun Choi[6*] and Seung Hyun Oh[1*]

Affiliations

[1]Gachon Institute of Pharmaceutical Sciences, Gachon University, Incheon 406-840, Republic of Korea

[2]National Cancer Center, Goyang-si, Gyeonggi-do 410-769, Republic of Korea

[3]Department of Zoology, School of Biological Sciences, Dr. Harisingh Gour Central University, Sagar 470003, India

[4]Korea Institute of Science and Technology (KIST), Gangneung Institute, Gangneung-si, Gangwon-do 210-340, Republic of Korea

[5]College of Korean Medicine, Daejeon University, Daejeon, 300-716, Republic of Korea.

[6]Department of Food and Nutrition, School of Food Science and Technology, Chung-Ang University, Anseong-si, Gyeonggi-do 456-756, Republic of Korea.

Correspondence

Dr. SH Oh, College of Pharmacy, Gachon University of Medicine and Science, Incheon, Republic of Korea

E-mail: eyeball@hanmail.net; Phone: + 82-32-899-6058; Fax: +82-32-899-6039

[*]Contributed equally**Abstract**

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Aim of the study: In this study, we examined the effect of different fractions and components of Chaga mushroom *(Inonotus Obliquus)* on viability and apoptosis of colon cancer cells. Among them, one component showed the most effective growth inhibition and was identified as ergosterol peroxide by NMR analysis. We investigated the anti-proliferative and apoptosis mechanisms of ergosterol peroxide associated with its anticancer activities in human colorectal cancer (CRC) cell lines and tested its anti-tumor effect on colitis-induced CRC developed by Azoxymethane (AOM) / Dextran sulfate sodium (DSS) in a mouse model.

Materials and methods: We used MTT (3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide) assays, flow cytometry assays, Western blot analysis, colony formation assays, reverse transcription-polymerase chain reaction (RT-PCR), immunohistochemistry (IHC), and AOM/DSS mouse models to study the molecular mechanism of metastatic activities in CRC cells.

Results: Ergosterol peroxide inhibited cell proliferation and also suppressed clonogenic colony formation in HCT116, HT-29, SW620 and DLD-1 CRC cell lines. The growth inhibition observed in these CRC cell lines was the result of apoptosis, which was confirmed by FACS analysis and Western blotting. Ergosterol peroxide inhibited the nuclear levels of β-catenin, which ultimately resulted in reduced transcription of c-Myc, cyclin D1, and CDK-8. Ergosterol peroxide administration showed a tendency to suppress tumor growth in the colon of AOM/DSS-treated mice, and quantification of the IHC staining showed a dramatic decrease in the Ki67-positive staining and an increase in the TUNEL staining of colonic epithelial cells in AOM/DSS-treated mice by ergosterol peroxide for both prevention and therapy.

Conclusion: Our data suggest that ergosterol peroxide suppresses the proliferation of CRC cell lines and effectively inhibits colitis-associated colon cancer in AOM/DSS-treated mice. Ergosterol peroxide down-regulated β-catenin signaling, which exerted anti-proliferative and pro-apoptotic activities in CRC cells. These properties of ergosterol peroxide advocate its use as a supplement in colon cancer chemoprevention.

Key Words

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Ergosterol peroxide, Colorectal cancer, AOM/DSS, β-catenin, Chaga Mushroom

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Introduction

People with inflammatory bowel diseases (IBDs), including ulcerative colitis (UC) and Crohn's disease (CD), are at increased risk for colorectal cancer (CRC). CRC accounts for approximately 15% of all deaths in IBD patients (Munkholm, 2003). CRC is the third most common cancer and the second cause of cancer-related death in the western world (Ferlay et al., 2010). Most CRC cases are sporadic and nonhereditary, accounting for approximately 75% of all incidences (Haggar and Boushey, 2009). While many genetic mutations occur during the multistep pathogenesis of CRC, among the earliest and most common events is the failure of the adenomatous polyposis coli (APC) tumor suppressor gene function (De Filippo et al., 2002). APC acts as a tumor suppressor by composing proteolytic destruction complex, which negatively regulates the Wnt signaling pathway (Stamos and Weis, 2013) by degradation of β-catenin, a transcriptional coactivator of the Wnt signaling pathway (MacDonald et al., 2009). Consequently, loss of APC stabilizes β-catenin and allows its subsequent translocation to the nucleus and activation of Wnt target genes by associating with LEF-1/TCF proteins (Clevers and Nusse, 2012; Stamos and Weis, 2013). Sustained Wnt signaling leads to the accumulation of transcription factors related to cell cycle progress and cancer (Polakis, 2012).

Drug development studies of active natural products or synthetic drugs for the treatment of colorectal cancer are ongoing for the inhibition or prevention of CRCs. Natural products have been used worldwide as traditional medicines against many diseases including cancer (Reddy et al., 2003). There are effective natural products available that have been used for medicine and health, exhibiting a wide range of biological activities such as anti-inflammatory, anti-oxidant, anti-mutagenic, and anti-cancer effects (Newman and Cragg, 2012; Rajamanickam and Agarwal, 2008).

Recently, the demand for more effective and safer therapeutic agents for the chemoprevention of human cancer has increased. Therefore natural products such as vegetables, fruits, plant extracts, and herbs have been a mainstay of cancer chemotherapy (Mann, 2002; Reddy et al., 2003). This will ensure that natural products make many future contributions to cancer chemotherapy.

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For many years, mushrooms have been used in traditional medicine to treat several diseases including many types of cancers. Chaga mushroom *(Inonotus Obliquus)*, commonly known as Chaga, has been used as a folk medicine in Russia and western Siberia for the prophylaxis and treatment of gastric disorders and even cancer (Kim et al., 2006). Chaga is claimed to have multiple natural beneficial properties for human health such as anti-bacterial, anti-allergic, anti-inflammatory, and anti-oxidant based on its bioactive compounds (Nagajyothi et al., 2014).

Recent studies have reported that a hot-water extract of *Inonotus obliquus* inhibits the proliferation of human colorectal cancer cells (HT-29) (Lee et al., 2009). In addition, the antitumor effect of three compounds (3β-hydroxy-lanosta-8,24-dien-21-al, inotodiol, and lanosterol) from *Inonotus obliquus* extracts were reported against Sarcoma-180-cell-bearing mice (Chung et al., 2010). We previously demonstrated that an aqueous extract of IOAE down-regulated β-catenin and NF-κB signaling, which exerted anti-proliferative and anti-inflammatory activities in colon cancer cells (Mishra et al., 2012). Wnt/β-catenin pathway plays an important role in cancers. More than 90% of colorectal cancers bear mutations that result in Wnt/β-catenin pathway activation, promoting their growth and survival. Although studies of the Chaga mushroom showed excellent anti-cancer activities against colon cancers, the bioactive compounds exhibiting those functions remain unknown. Therefore we isolated several bioactive compounds from Chaga mushroom and analyzed the anti-cancer activities.

In this study, we examined the effect of different fractions and components of Chaga mushroom on viability and apoptosis of colon cancer cells. We analyzed the effects of ergosterol peroxide on human CRC cell lines and on colitis-induced CRC developed by Azoxymethane/Dextran sulfate sodium (AOM/DSS) in a mouse model. We investigated the anti-proliferative and apoptosis mechanisms of ergosterol peroxide associated with its anticancer activities against intestinal tumorigenesis.

Materials and Methods

Animals, Cells and materials

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Female 6-week-old C57BL/6 mice were purchased from Jackson Laboratory (Bar Harbor, ME, USA). HCT116, DLD-1, HT-29 and SW620 were obtained from the American Type Culture Collection (Manassas, VA, USA). Colon cancer cells were cultured in RPMI1640 medium supplemented with 10% fetal bovine serum (FBS) and penicillin/streptomycin (GIBCO Invitrogen, Grand Island, NY, USA). Azoxymethane (AOM) was purchased from Sigma-Aldrich (St. Louis, MO, USA) and dextran sodium sulfate (DSS) was purchased from MP Biomedicals (Santa Ana, CA, USA). Purified ergosterol peroxide from Chaga mushroom was dissolved in dimethyl sulfoxide (DMSO) and the cells were treated with the indicated dose of ergosterol peroxide or 0.1% DMSO (control). Ergosterol peroxide for animal experiment was synthesized in Dongyoon Shin's laboratory. Antibodies against PARP, caspase-3, β-catenin, fibrillarin, cyclin D1, CDK-8 and tubulin were purchased from Cell Signaling Technology (Danvers, MA, USA), and those against actin and c-Myc were purchased from Santa Cruz Biotechnology (Santa Cruz, CA, USA).

Identification of fraction 15 (Ergosterol peroxide)

Chaga mushroom was purchased from K&T Limited Company (Biysk, Russia). The dried Chaga mushrooms (8 kg) were extracted with 95% ethanol three times under reflux for 4 hr. The extract was filtered and evaporated in vacuo to yield the ethanol extract (212.95 g). This ethanol extract was then suspended in distilled water and sequentially partitioned with n-hexane, ethyl acetate, and n-butanol. The *n*-hexane fraction (28 g) was chromatographed on a silica gel column using Hexane-EtOAc gradient system (30:1→ 15:1→ 8:1→ 4:1→ 1:1, v/v, 2 L) to give eleven fractions (fraction 1-11). Fractions 1-10 were obtained by eluting with 1 L of solvent. Fraction 11 was eluted with Hexane-EtOAc (0:100, v/v, 2 L). Fraction15 (360 mg, white crystals) was finally purified by re–crystallization of fraction 10 (650 mg). Identification of Fraction15 was carried out by NMR analysis. Chemical structure of fraction 15

00000000. was confirmed by ^{1}H NMR and ^{13}C NMR data compared with reported data in literature (Kim et al., 2011).

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MTT assay

The inhibitory effects of an aqueous extract of *Inonotus obliquus* (IOAE), various fractions from Chaga mushroom, and ergosterol peroxide on the growth of CRC cell lines were determined using a 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide (MTT, Sigma-Aldrich) assay. CRC cells were plated in 96-well plates (3×10^3 cells/well) and cultured in medium containing various concentrations (5–20 μg/mL) of the samples. The cells were allowed to grow for a total of 48 hr. After addition of 10 μl MTT (5 mg/mL) and further incubation at 37°C for 2 hr, the medium was removed and 100 μl DMSO was added to each well. The optical density of each well at 562 nm was measured using a microplate spectrometer. Three independent experiments were performed with similar results; representative results of one experiment are presented.

Cell cycle arrest and apoptosis analysis

The effect of ergosterol peroxide on cell cycle arrest and apoptosis in CRC cells was determined using flow cytometric analysis. Briefly, 2×10^5 cells were seeded into 60-mm dishes. After 24 hr, cells were treated with ergosterol peroxide (20 μg/mL) for 48 hr. Cells were then trypsinized, rinsed once, re-suspended in phosphate-buffered saline (PBS), fixed with ice-cold 70% ethanol for at least 20 minutes and stored at −20°C. Fixed cells were rinsed again and re-suspended in PBS containing 50 μg/mL propidium iodide (Sigma, St. Louis, MO, USA) and 40 KU/mL DNase-free RNase (Stratagene, La Jolla, CA, USA). Apoptotic cells were stained with Annexin V-fluorescein isothiocyanate (FITC) and propidium iodide (BD Biosciences, San Jose, CA, USA). Briefly, 5×10^5 cells were washed with PBS and then labeled with Annexin V–FITC and propidium iodide as instructed by the manufacturer. Flow cytometric analysis was performed using a BD FACS Calibur flow cytometer (BD) equipped with a 488-nm argon-ion laser. Approximately 10,000 cells were evaluated for each sample. Three independent experiments were performed with similar results; representative results of one experiment are presented.

Western blot analysis

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Total cell extracts of human CRC cells were treated with ergosterol peroxide (5–20 μg/mL) as indicated. Preparation of whole-cell lysates, protein quantification, gel electrophoresis, and Western blotting were performed as described previously (Kang et al., 2011). Protein concentrations were measured using a bicinchoninic acid protein assay (Pierce Biotechnology, Rockford, IL, USA). Equal amounts of protein from cell lysates from each treatment group were resolved using sodium dodecyl sulfate-polyacrylamide gel electrophoresis and immunoblotted with primary antibodies. Antibody binding to the membrane was detected using enhanced chemiluminescence Western blotting detection reagents obtained from GE Healthcare (Chalfont St. Giles, UK). For nuclear protein isolation, cells were homogenized with lysis buffer (10 mM Tris HCl pH 7.5 with 2 mM $MgCl_2$, 3 mM $CaCl_2$, 0.5 M sucrose and 1x protease and phosphatase inhibitor cocktails) and then mixed with Igepal CA-360 solution (0.6%). Cell lysate was centrifuged and the supernatant containing cytoplasmic fraction was collected. The pellet was treated with extraction buffer (20 mM HEPES buffer pH 7.9 with 1.5 mM $MgCl_2$, 0.42 M NaCl, 0.2 mM EDTA, 25% (v/v) glycerol, 0.1 M DTT, and 1x protease and phosphatase inhibitor cocktails) and then centrifuged to obtain supernatant containing nuclear fraction.

Anchorage-dependent and –independent colony formation assay

The effect of ergosterol peroxide on CRC cell colony formation was tested using a colony formation assay. For anchorage-dependent assays, cells were plated overnight in 6-well plates at a density of 5×10^2 cells/well and treated with the indicated concentrations of ergosterol peroxide for 2 weeks. The cells were then fixed with methanol and stained with crystal violet. A soft agar colony formation assay was performed to assess anchorage-independent growth of CRC cells treated with ergosterol peroxide. RPMI1640 containing 10% FBS and 1% agarose was placed in 12-well plates as the base agar. Cells were seeded at a density of 1×10^3 cells/well in top agar, which consisted of RPMI1640 with 10% FBS containing 0.5% agarose. The plates were then incubated at 37°C in a humidified incubator for 3 weeks. During this time, cells were treated twice a week with cell culture media containing 5 or 20 μg/mL ergosterol peroxide. The colonies measuring ≥30 μm were counted

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using a dissection microscope (Woo et al., 2015). Three independent experiments were performed with similar results; representative results of one experiment are presented.

Colitis-induced adenocarcinoma mouse models

All animal procedures were conducted in accordance with a protocol approved by the Institutional Animal Care and Usage Committee (IACUC) at Gachon University in Incheon, Korea. Female 5-week-old C57BL/6 mice were purchased from Charles River Laboratories (Wilmington, MA, USA). Animals were randomly divided into four groups (n=15) (Fig. 4A). Colitis-induced colon adenocarcinoma was induced by single-dose, intraperitoneal (i.p.) injection of AOM (10 mg/kg) (Sigma Chemical Co.) followed by colonic inflammation induced by 2% (w/v) DSS (MW; 36,000-50,000, MP Biomedical, Solon, OH) in the drinking water for 7 d (groups 2, 3 and 4). AOM is an indirect and genotoxic carcinogen requiring metabolic activation of CYP2E1 to exert its carcinogenic action (Sohn et al., 2001). Mice in two groups were administered ergosterol peroxide (15 mg/kg) dissolved in vehicle (DMSO : Cremophor : water = 1: 1 : 8) twice a day by oral gavage for 14 weeks (prevention, group 3) or 8 weeks (therapy, group 4). The body weight of mice was recorded every week. Mice were euthanized by CO_2 inhalation and colons were excised. Histopathologic evidence of toxicity and injury in other organs, such as liver, kidney, and spleen were evaluated by a veterinary pathologist.

Histological analysis

The formalin-fixed colon tissues were processed, embedded in paraffin, and used for hematoxylin and eosin (H&E) staining. Colons from mice were prepared as a 'swiss roll' for full-length histopathology work. Tissues in paraffin block were sectioned at 4-μm thickness. Frozen colon tissue sections were reacted with diluted primary antibodies for cyclin D1 (1:50), c-Myc (1:50), β-catenin (1:100) and Ki67 (1:1,000), and incubated with biotinylated goat anti-rabbit secondary antibody and avidin-biotin peroxidase. Sections were visualized by chromogen 3,3'-diaminobenzidine (Dako, K3466) and counterstained with hematoxylin (Woo et al., 2015). The intensity and localization

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of immunoreactivity against antibodies was viewed under a light microscope (200×) by Aperio ScanScope (Aperio Technologies, CA, USA). Cell proliferation was evaluated by staining for Ki67 nuclear antigen, and apoptosis was evaluated by staining for terminal deoxynucleotidyl transferase dUTP nick end labelling (TUNEL) using frozen colon tissue slides. The TUNEL assay was performed as previously described (Oh et al., 2007). For apoptotic analysis, TUNEL-positive cells were counted in 3-5 randomly selected fields of view, at a magnification of x400.

Reverse transcription-polymerase chain reaction (RT-PCR)

To measure gene expression changes at the RNA level, reverse transcriptase (RT)-polymerase chain reaction (PCR) was performed. Total RNA was isolated from CRC cells using TRIzol reagent (Invitrogen, Carlsbad, CA). First, complementary DNA was synthesized from 2 μg of RNA extracted from the cells using an M-MLV RT kit (Invitrogen). RT-PCR was carried out with gene-specific primers for β-catenin (forward, 5'- GCG TGG ACA ATG GCT ACT CA-3'; reverse, 5'- GAG TTG TAA TGG CAT AAA ACA AC-3'). Primers amplifying a region of β-actin (forward, 5'- GTG GGG CGC CCC AGG CAC CA-3'; reverse, 5'- CTC CTT AAT GTC ACG CAC GAT TTC-3') were used as internal controls. After PCR, the products were analyzed by separation on a 1.5% agarose gel in tris-acetate/ethylenediaminetetraacetic acid (EDTA) buffer.

Statistical Analysis

Data are shown as means ± standard deviations. All statistical analysis was performed using One-way analysis of variance (One-way ANOVA). For post hoc test, Scheffe was used. The significance level was limited to 5% ($p < 0.05$). SPSS 22.0 was used as a statistical analysis tool.

Results

Inhibition of Cell Proliferation by Ergosterol Peroxide.

Previously we showed anti-cancer effect of Chaga mushroom (*Inonotus obliquus*) in colon adenocarcinoma *in vitro* and *in vivo* (Mishra et al., 2013). Therefore we searched for the active

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components of Chaga mushroom for anti-cancer therapy. Among 4 fractions from different solvents (ethanol, n-hexane, ethyl acetate, and n-butanol), the hexane-soluble fraction (IOH), showed the greatest anti-proliferative effect compared to aqueous extract of *Inonotus obliquus* (IOAE) on all cancer cells tested in this experiment (Supplementary Table 1). To further determine the effective components, IOH was chromatographed and fractioned into 15 fractions as described above. CRC cells were treated with these 15 fractions and the anti-proliferative effects were compared with that of IOAE. Fraction 15 showed the most effective growth inhibition in the all of the CRC cells used in the experiment (Supplementary Table 2). We identified fraction 15 as ergosterol peroxide by NMR analysis (Supplementary Fig. 1).

We treated HCT116, HT-29, SW620 and DLD-1 cells with the indicated concentrations of ergosterol peroxide for 48 hr and confirmed that ergosterol peroxide inhibited CRC cell proliferation in a concentration-dependent manner in all cell lines (Fig. 1A). We further investigated whether ergosterol peroxide from Chaga mushroom suppresses colony formation in CRC cells. Interestingly, long-term treatment with ergosterol peroxide (at concentrations of 10 μg/mL in assays for anchorage-dependent colony formation and 5 μg/mL in assays for anchorage-independent colony formation) significantly reduced both anchorage-dependent and -independent colony formation in HCT116, HT-29, SW620 and DLD-1 cells (Figs. 1B and 1C), further confirming the anticancer effects of ergosterol peroxide. Of note, 5 μg/mL of ergosterol peroxide completely inhibited anchorage-dependent colony formation in HT-29 cells,

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Effect of Ergosterol Peroxide on Cell-Death and Apoptosis in CRC Cells.

To determine whether the growth inhibition observed in these CRC cell lines was the result of apoptosis, we next analyzed cell-cycle arrest and apoptosis. As shown in Fig. 2A, ergosterol peroxide induced subG1 arrest in all CRC cell lines used in this experiment. FACS analysis after annexin V and propidium iodide staining showed that treatment with 20 μg/mL ergosterol peroxide induced apoptosis (annexin V-positive or propidium iodide-positive cells) in HCT116 and HT-29 cancer cells (Fig. 2B). To evaluate the pro-apoptotic response of ergosterol peroxide, we investigated the effect of

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ergosterol peroxide on expression of poly (ADP-ribose) polymerase (PARP) and caspase-3. Caspase-3 activation (decrease of uncleaved caspase-3) by ergosterol peroxide was followed by cleavage of PARP, with a marked increase in cleaved PARP in colon cells (Fig. 2C). These results suggest that ergosterol peroxide induces cell-cycle arrest and apoptosis.

Ergosterol Peroxide Down-regulates β-catenin Activation in Colon Cancer

We previously demonstrated that an aqueous extract from Chaga mushroom exerted anti-proliferative and anti-inflammatory activities in colon cancer cells by down-regulating β-catenin (Mishra et al., 2012). To investigate the effect of ergosterol peroxide on the β-catenin pathway in CRC cells, cells were treated with ergosterol peroxide and Western blotting was performed. We observed that treatment with ergosterol peroxide reduced β-catenin expression in all of the cells (Fig. 3A, left). To examine whether the decreased expression of β-catenin was mediated by mRNA expression, RT-PCR analysis of β-catenin was performed in HCT116 and HT-29 cells. Interestingly, ergosterol peroxide treatments did not change β-catenin mRNA expression levels in HCT116 and HT-29 cells (Fig. 3A, right). Next, we analyzed the protein levels of β-catenin in cytosolic and nuclear fractions of ergosterol peroxide-treated HCT116 and HT-29 cells. As shown in Fig. 3B, ergosterol peroxide completely suppressed the levels of β-catenin in nuclear lysates of HCT116 cells and HT-29 cells at 20μg/mL and 5μg/mL, respectively, but it did not change β-catenin expression in the cytoplasmic fraction (Fig. 3B). Consequently ergosterol peroxide suppressed the levels of c-Myc, cyclin D1 and cyclin-dependent protein kinase 8 (CDK-8) (down-stream targets of β-catenin) in a dose-dependent manner in CRC cell lines (Fig. 3C). These results demonstrate that ergosterol peroxide inhibits the nuclear levels of β-catenin, which ultimately results in reduced transcription of c-Myc, cyclin D1 and CDK-8 (Fig. 3C).

Ergosterol Peroxide Suppresses Tumor Development in AOM/DSS-Treated Mice

To investigate the effect of ergosterol peroxide on an established colon cancer model, we used the AOM/DSS-induced adenocarcinoma mouse model (Fig. 4A). Colonic inflammation and mucosal

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injury in AOM/DSS-treated mice was analyzed by histological examination of H&E-stained colon tissues (Fig. 4B). Tissue sections from representative areas of colon of control mice showed intact surface epithelium, cryptal gland, stroma and submucosa, while AOM/DSS treatment showed distorted crypt epithelium, extensive mucosal damage, and a large number of inflammatory cells in the microscopic analysis. Ergosterol peroxide treatment showed relatively well-preserved crypt structures and improved histological features (Fig. 4B). Ergosterol peroxide treatment did not affect tumor incidence in AOM/DSS-treated mice (Fig. 4C). However, ergosterol peroxide administration tended to suppress tumor growth in the colon of AOM/DSS-treated mice (Fig. 4C, left), resulting in a significant decrease in total tumor count in both prevention ($p = 0.006$) and therapy ($p = 0.009$) groups (Fig. 4C, right). Further, we measured the size of tumors under dissecting microscope and classified them into two categories: small (<2 mm) and large (>2mm). Total count of large tumors was reduced by ergosterol peroxide treatment in prevention ($p = 0.015$) and therapy ($p = 0.067$) groups (Fig. 4C). Also, total counts of small tumors were also reduced by ergosterol peroxide treatment in prevention ($p = 0.097$) and therapy ($p = 0.043$) groups (Fig. 4C). Next, we assessed the proliferation and survival of intestinal epithelial cells in AOM/DSS-treated mice by IHC staining of cellular proliferation marker Ki67. The results showed an aberrant increase in the proliferation of colon crypt cells in AOM/DSS-treated mice (Fig. 4D). Treatment with ergosterol peroxide for both prevention and therapy caused a dramatic decrease in the Ki67-positive staining of colonic epithelial cells in AOM/DSS-treated mice (Fig. 4D). These results suggest that the proportion of actively proliferating cells was very high in the colon of AOM/DSS-treated mice suppressed by ergosterol peroxide. Conversely, TUNEL staining showed that treatment with ergosterol peroxide for both prevention and therapy increased apoptosis of colonic epithelial cells compared to AOM/DSS-treated mice (Fig. 4D). Collectively, these results demonstrate that ergosterol peroxide helps maintain colonic epithelial cell structures, improves histological damage in response to AOM/DSS, and suppresses tumor growth in the colon.

Ergosterol Peroxide Down-regulates β-catenin Activation in Colon Cancer Model

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Next, we investigated the effect of ergosterol peroxide on in situ expression of β-catenin. IHC results showed increased staining of β-catenin in intestinal epithelial cells and increased number of β-catenin-positive intestinal epithelial cells of AOM/DSS-treated mice (Fig. 5). Overexpression of β-catenin shows highly proliferating cellular milieu in the intestinal epithelium of AOM/DSS-treated mice and was suppressed by ergosterol peroxide. Also the expression of c-Myc and cyclin D1 was increased in the intestinal epithelium of AOM/DSS-treated mice and it was decreased by ergosterol peroxide. Again, these results demonstrate that ergosterol peroxide helps maintain colonic epithelial cell structures, improves histological damage in response to AOM/DSS, and suppresses tumor growth in the colon.

Discussion

We previously demonstrated that an aqueous extract of *Inonotus obliquus* (IOAE) showed potent anti-inflammatory activity by down-regulating the intricate Wnt/ β-catenin signaling and NF-κB pathway *in vitro* and *in vivo* without any toxicity (Mishra et al., 2013). To understand which bioactive compounds exhibit those functions, we isolated and analyzed several bioactive compounds from Chaga mushroom and studied the anti-cancer activities for cancer therapy.

In this research, we fractionated Chaga mushroom by different solvents and tested the anti-proliferative effect of each fraction compared to IOAE. Interestingly, fraction 15 showed most effective growth inhibition in the all the tested CRC cell lines. We identified fraction 15 as ergosterol peroxide (EP) by NMR analysis (Supplementary Fig. 1). Ergosterol peroxide has been isolated from various species of edible and medicinal mushrooms (Rhee et al., 2012). Recent studies have reported that ergosterol peroxide suppresses LPS-induced pro-inflammatory gene expression in RAW264.7 macrophage-like cells and the growth of HT-29 colon adenocarcinoma cells (Kobori et al., 2007). Ergosterol peroxide and (22E)-ergosta-7,22-dien-5α-hydroxy-3,6-dione attenuated the growth of prostate cells, at least in part, by triggering an apoptotic process (Russo et al., 2010). Recently ergosterol peroxide was shown to have anti-tumor activity in multiple myeloma U266 cells with anti-angiogenic activity targeting the JAK2/STAT3 signaling pathway (Rhee et al., 2012). Recent studies have reported that a hot-water extract of *Inonotus obliquus* inhibits the proliferation of human

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colorectal cancer cells (HT-29) (Lee et al., 2009). In addition, the antitumor effect of three compounds (3β-hydroxy-lanosta-8,24-dien-21-al, inotodiol, and lanosterol) from *Inonotus obliquus* extracts were reported against Sarcoma-180-cell-bearing mice (Chung et al., 2010).

Ergosterol peroxide inhibited not only cell proliferation but also suppressed clonogenic colony formation in HCT116, HT-29, SW620 and DLD-1 CRC cell lines with different susceptibility. The growth inhibition observed in these CRC cell lines was the result of apoptosis, which was confirmed by FACS analysis and Western blotting, consistent with a previous report that ergosterol peroxide inhibited growth and induced apoptosis of HL60 cells (Takei et al., 2005). Ergosterol peroxide inhibited the nuclear levels of β-catenin, which ultimately resulted in reduced transcription of c-Myc, cyclin D1, and CDK-8. c-Myc, cyclin D1, and CDK-8 are important down-stream targets of β-catenin that regulate malignancies including cell cycle and apoptosis in cancer (Kaur et al., 2010). Next, we applied ergosterol peroxide to an animal model for evaluation as a cancer therapy agent. We used the AOM/DSS-induced adenocarcinoma mouse model, in which tumors only develop in the colon through multi-step carcinogenesis that reflects the entire progress of tumor growth in man (Yoshimi et al., 2012). Ergosterol peroxide administration showed a tendency to suppress tumor growth in the colon of AOM/DSS-treated mice, and quantification of the IHC staining showed a dramatic decrease in the Ki67-positive staining and an increase in the TUNEL staining of colonic epithelial cells in AOM/DSS-treated mice by ergosterol peroxide as both prevention and therapy. However, we saw no beneficial effect in the prevention group compared to the therapy group in suppressing large-sized tumor in AOM/DSS-treated mice (Fig. 4C). When toxic effects on other organs were evaluated by a veterinary pathologist in AOM/DSS-treated mice, ergosterol peroxide was well tolerated, with no detectable organ toxic effects after a 8-week or 14-week treatment to achieve therapeutic or chemopreventive activities (data not shown).

We, for the first time, suggest ergosterol peroxide from Chaga mushroom as a chemotherapeutic agent. Our data show that ergosterol peroxide suppresses the proliferation of CRC cell lines and effectively inhibits colitis-associated colon cancer in AOM/DSS-treated mice models. Ergosterol peroxide down-regulates β-catenin signaling, which exerts anti-proliferative and pro-

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apoptotic activities in CRC cells. These properties of ergosterol peroxide advocate its use as a supplement in colon cancer chemoprevention.

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Legends for Figures

Fig. 1 Inhibition of cancer cell proliferation by ergosterol peroxide. (a) Proliferation of HCT116, HT-29, SW620 and DLD-1 CRC cells after 48 hr treatment with ergosterol peroxide at the indicated concentrations. (b) Anchorage-dependent colony formation assay of CRC cells following treatment with ergosterol peroxide for 2 weeks. (c) Anchorage-independent colony formation assay of colon cancer cells following treatment with ergosterol peroxide for 3 weeks. EP, ergosterol peroxide. The values shown are means ± SD. *P<0.05; **P<0.01; ***P<0.001 compared with the untreated group (One-way ANOVA).

Fig. 2 Effect of ergosterol peroxide on cell death and apoptosis in colorectal cancer cells. (a) HCT116, HT-29, SW620 and DLD-1 colorectal cancer cells were treated with ergosterol peroxide (20 μg/mL) for 48 hr. Cells were stained with propidium iodide for cell cycle analysis using flow cytometry. (b) Apoptosis and/or necrosis in colorectal cancer cells was evaluated using annexin V-FITC and propidium iodide staining, and ergosterol peroxide-induced cell death was analyzed using flow cytometry. (c) Western blot analysis of CRC cells following treatment with ergosterol peroxide for 48 hr. EP, ergosterol peroxide.

Fig. 3 Ergosterol peroxide down-regulates β-Catenin activation in colorectal cancer cells. (a) HCT116, HT-29 and SW620 colorectal cancer cells were treated with ergosterol peroxide (5 or 20 μg/mL) for 48 hr. Protein and mRNA levels of β-catenin were tested by western blotting and RT-PCR. (b) HCT116, HT-29 colorectal cancer cells were treated with ergosterol peroxide (5 or 20 μg/mL) for 48 hr. Nuclear and cytoplasmic fractions were separated and western blot was performed for analysis of β-catenin. Fibrillarin and α-tubulin were used as loading controls for nuclear and cytoplasmic

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fractions, respectively. (c) Western blot analysis of colorectal cancer cells following treatment with ergosterol peroxide for 48 hr. c-Myc, cyclin D1, and cyclin-dependent protein kinase 8 (CDK-8), which are down-stream targets of β-catenin, were tested. EP, ergosterol peroxide.

Fig. 4 Ergosterol peroxide suppresses tumor development in AOM/DSS-induced adenocarcinoma model. (a) Experimental protocol used for carcinoma induction by AOM (10mg/kg) and DSS (2% DSS) in female C57BL/6 mice. Mice were treated with either vehicle control or ergosterol peroxide (prevention:15 mg/kg or therapy:15 mg/kg) twice a day by oral gavage. (b) Representative images of H&E-stained colon sections, visualized under the light microscope. Slides were scanned using an Aperio ScanScope. Arrow indicates muscularis mucosa layer and star indicates lumen of intestine. Arrows indicate intact intestinal gland. Scale bar, 1 mm (upper) or 100 μm (lower). (c) The number of tumors observed in the colon of mice treated with AOM/DSS and/or ergosterol peroxide. Colon was excised from mice and total number of tumors was counted. The sizes of tumors were measured and classified as small (<2 mm) and large (>2mm). Data presented are means+SD values. *P<0.05; **P<0.01; ***P<0.001 compared with AOM/DSS-treated mouse group (One-way ANOVA). (d) Immunohistochemistry for Ki67 and TUNEL in AOM/DSS-treated mouse colon tissue visualized by light microscope and confocal microscope (x 400). Arrows indicate proliferating cells (upper) or apoptotic cells in intestinal epithelial cells or tumors. Scale bar, 50 μm.

Fig. 5 Ergosterol peroxide down-regulates β-Catenin activation in colon cancer model. Immunohistochemistry for β-catenin, c-Myc, and cyclin D1 in AOM/DSS-treated mouse frozen colon tissue visualized under the light microscope (x 400). Scale bar, 50 μm. EP, ergosterol peroxide.

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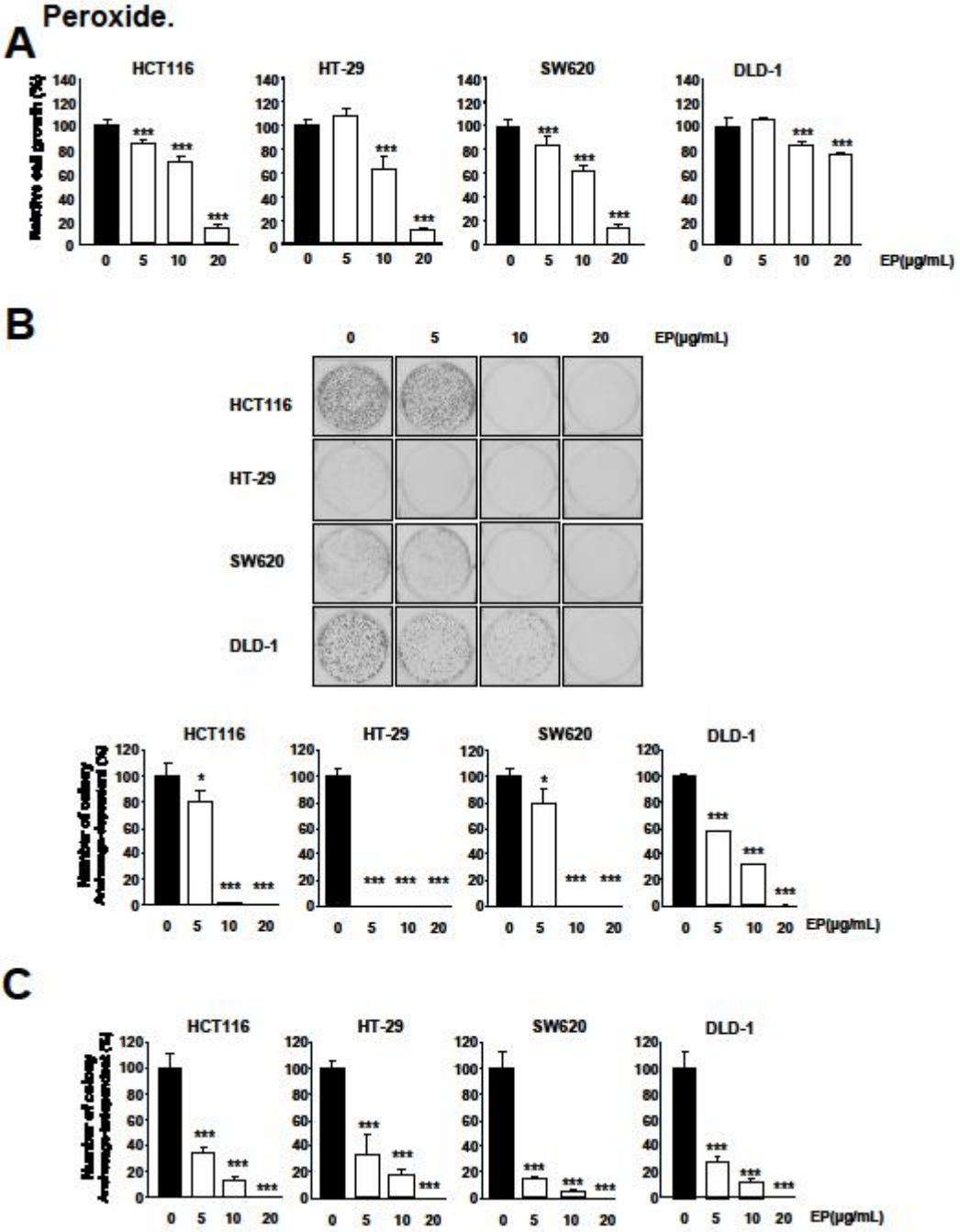
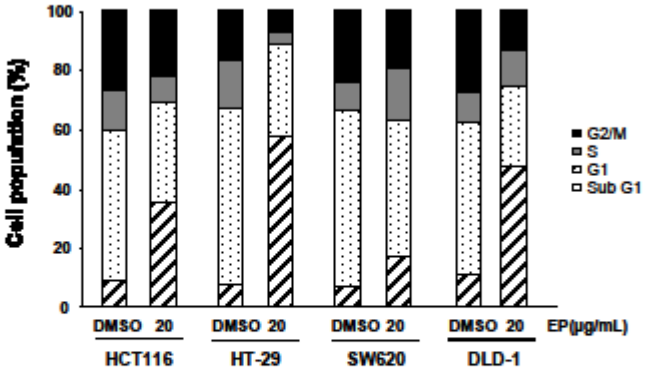
Inhibition of cancer cell proliferation by Ergosterol Peroxide. **Fig. 1**



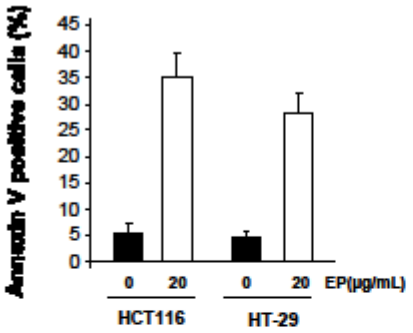
Effect of Ergosterol Peroxide on cell-death and apoptosis in CRC cells. **Fig. 2**

A



	HTC116		HT-29		SW620		DLD-1	
	DMSO	EP	DMSO	EP	DMSO	EP	DMSO	EP
sub-G1	5.85±1.92	25.05±1.57	6.80±1.68	55.08±4.88	6.33±2.76	14.76±1.72	7.98±1.80	27.70±3.39
G1	33.55±4.61	23.72±3.48	51.04±1.09	29.77±1.48	50.45±2.55	39.40±4.90	35.85±4.64	15.51±2.69
S	9.02±1.54	6.41±2.47	14.13±3.37	4.09±1.44	7.82±1.59	14.72±1.95	7.07±2.04	7.11±0.88
G2/M	17.88±2.62	15.63±0.92	14.67±3.36	6.87±1.47	20.58±4.00	16.66±3.26	19.34±2.01	7.84±4.32

B



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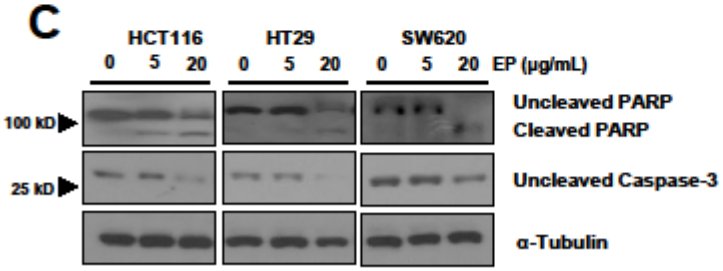
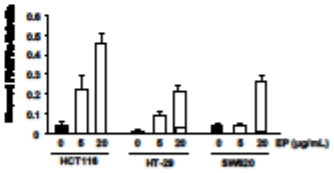
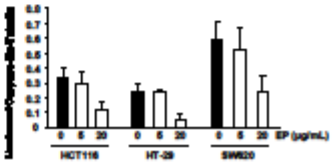
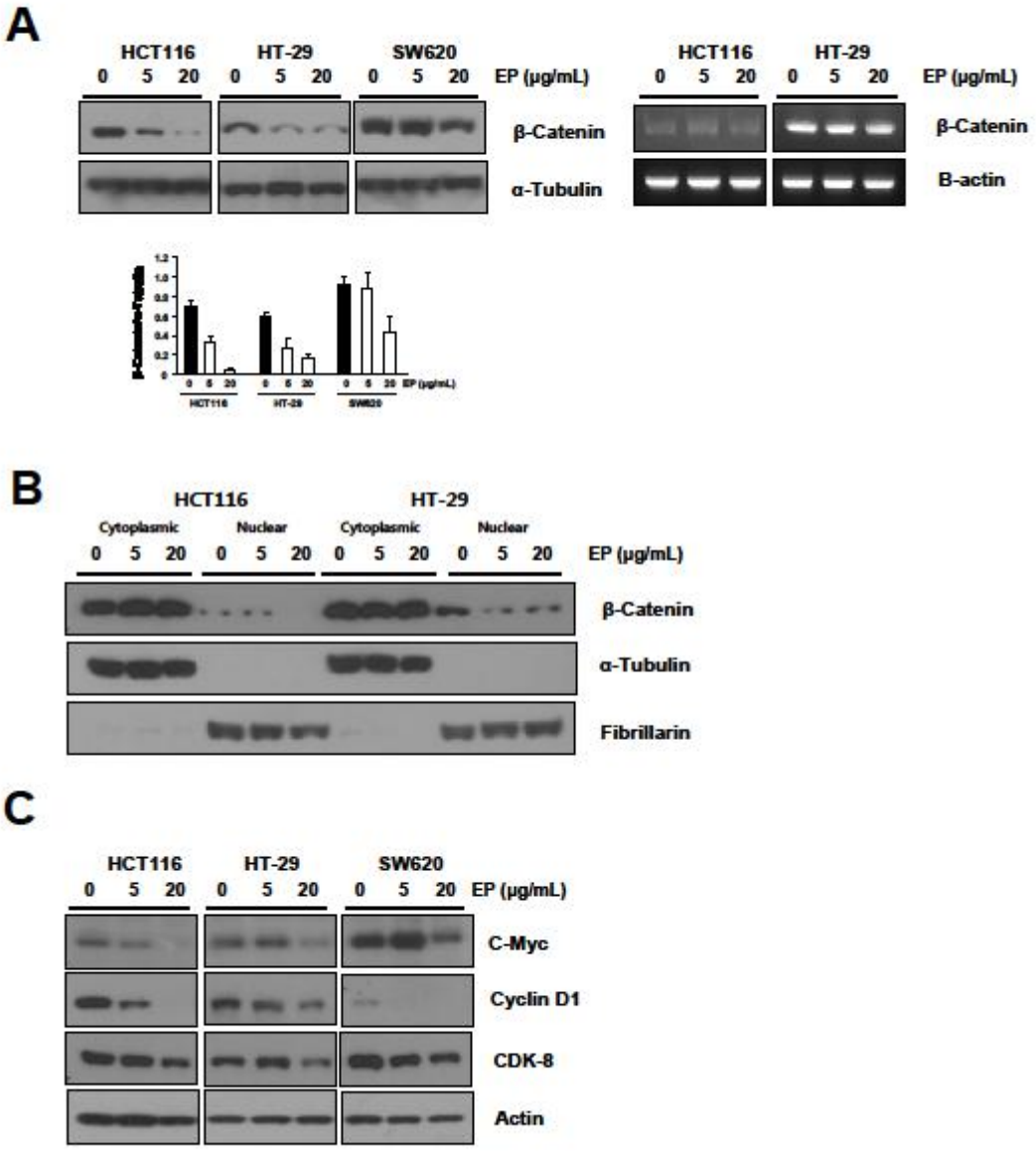






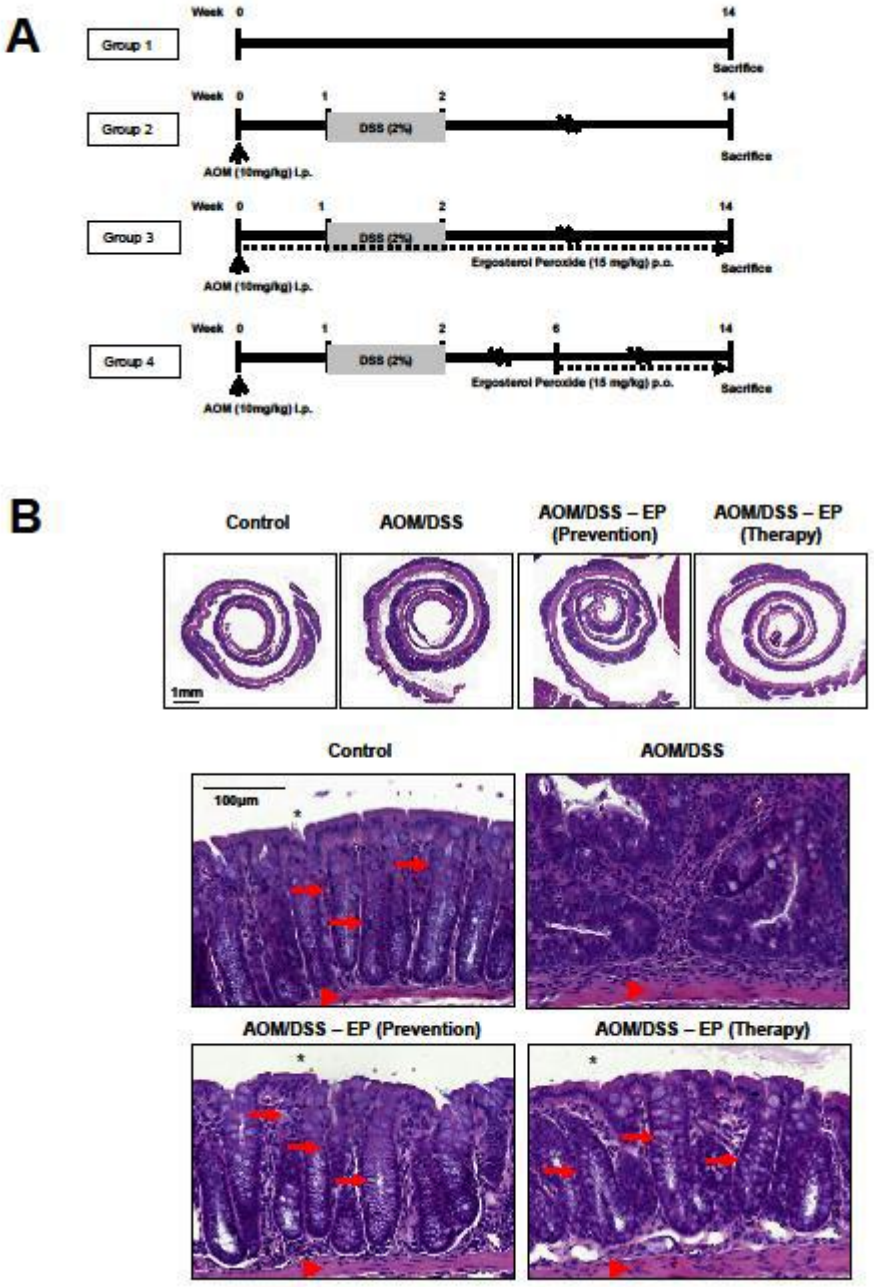
Ergosterol Peroxide Down-regulates β-Catenin Activation in CRC

Fig. 3

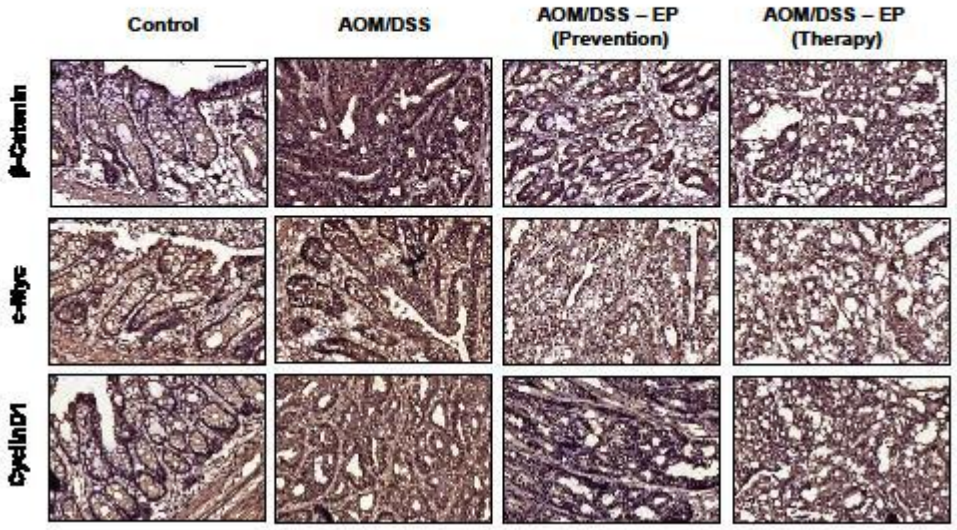


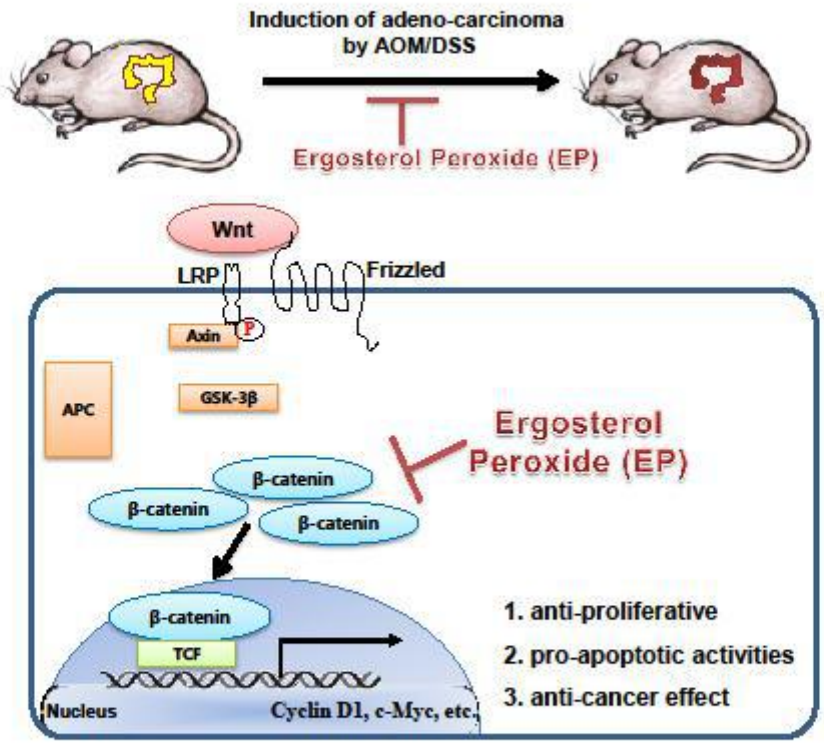
Ergosterol Peroxide Suppresses Tumor Development in AOM/DSS-Treated Mice

Fig. 4



Ergosterol Peroxide Down-regulates β-Catenin Activation in Colon Cancer model **Fig. 5**




Induction of adeno-carcinoma
by AOM/DSS
Ergosterol Peroxide (EP)
Wnt
LRP
Frizzled
Axin
P
APC
GSK-3β
Ergosterol
Peroxide (EP)
β-catenin
β-catenin
β-catenin
β-catenin
TCF
Nucleus
Cyclin D1, c-Myc, etc.
1. anti-proliferative
2. pro-apoptotic activities
3. anti-cancer effect

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